

Laxman K R

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AI Engineer specializing in autonomous multi-agent systems, microfrontend architecture, and system design.

WORK EXPERIENCE

Tata Consultancy Services

November 2024 – Present

AI Engineer

Chennai, TN

- Engineered an autonomous multi-agent AMS in Python that reduced MTTR by 63% through automated anomaly detection, incident deduplication, and SOP-assisted resolution using a vector-indexed knowledge base.
- Designed a Next.js dashboard with progressive autonomy from human-resolved to human-in-the-loop approval to fully autonomous operation, achieving 6-minute average response time via a Planning-Executor-Validator pipeline.
- Drove migration of 50+ modules from a monolithic Next.js app into 10 separate microfrontends, serving 20+ developers and reducing build time from 10 to 5 minutes while increasing deployment throughput by 2.5 times.
- Shipped a Python/FastAPI backend with Google Cloud Storage to serve 120+ global AI offerings as a unified catalog for Google Cloud Next 2026, with a Next.js frontend achieving 96% Lighthouse score and sub-second page loads.
- Managed CI/CD pipelines across 3 cross-functional teams (infra, sales, BA), delivering 25+ monthly deployments to production with zero critical rollbacks.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, SQL

AI & Agents: GenAI, AI/ML, LLMs, RAG, MCP, AI SDK, Google ADK, AutoGen, LangChain, LangGraph, Qdrant, Pinecone

Frameworks: React, Next.js, FastAPI, Node.js, Express, Effect TS, Bun

Databases & Infra: MongoDB, PostgreSQL, Redis, SQLite, AWS Lambda, Cloudflare, Kubernetes, Docker, Jenkins

Tools: Git, REST APIs

PROJECTS

Retail Platform

July 2024 – June 2026

Go, Next.js, TypeScript, SQLite, AWS Lambda, Cloudflare

GitHub

- Built the frontend with Next.js, delivering an admin dashboard, CMS, and interfaces for search and product recommendations serving 200+ products.
- Designed a decoupled microservices backend in Go with 7 services deployed via AWS Lambda, powering the search engine, recommendation engine, notification service, and order management through event-driven architecture.

Kite

March 2026 – April 2026

AG-UI, TanStack Start, React 19

Demo

- Built an AI assistant that renders UI components via AG-UI protocol, with an AI SDK agent orchestrating dynamic UI generation in a TanStack Start app.
- Streamed agent responses with structured UI rendering over the AG-UI protocol, built on a TanStack Start application following React composition patterns and the Async React standard.

Cortex

April 2026 – May 2026

Effect TS, TypeScript, Vector Databases

Demo

- Architected an Effect-native ORM for vector databases with structured concurrency, compositional error handling, and type-safe dependency injection; includes a query builder, schema migrations, and scheduling, achieving 500+ npm downloads.
- Implemented an adapter pattern for Zvec and PageIndex, enabling portable multi-tenant RAG pipelines across different vector database providers.

EDUCATION

Amrita Vishwa Vidyapeetham

September 2020 – May 2024

B.Tech in Computer Science and Engineering

Coimbatore, TN

- Research:** Complex or Irregular Lung Nodule Detection from CT Scans